Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	
Project Name	EV Market Intelligence Dashboard
Maximum Marks	4 Marks

Data Flow Diagrams:

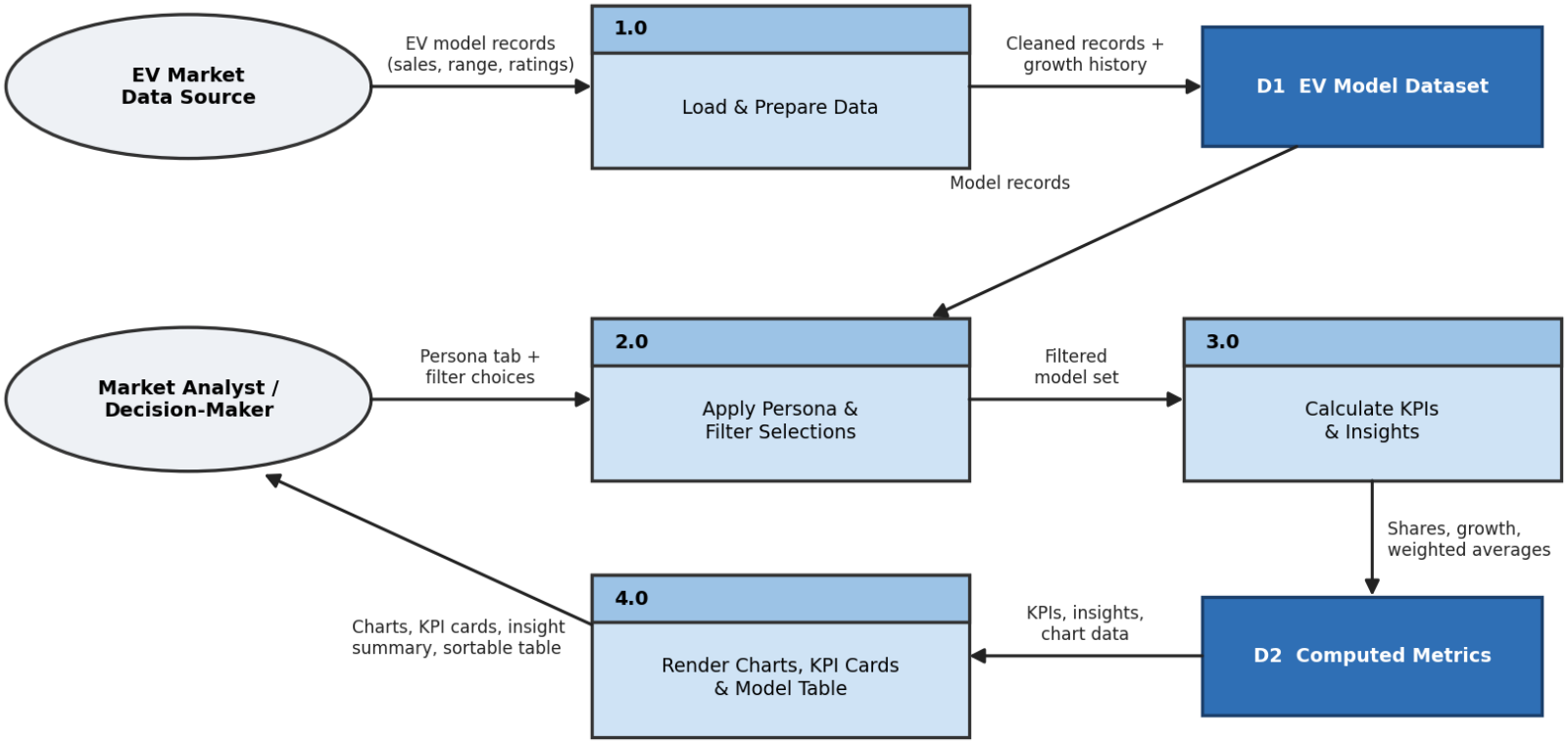
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).

Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

DFD Level 0 – EV Market Intelligence Dashboard



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Market Strategy Director (Emily Johnson)	Category & Segment Analysis	USN-1	As a Market Strategy Director, I can view sales by EV category so that I can see which categories drive sales.	Category chart and KPI cards show 2025 sales and share for each category	High	Sprint-1
		USN-2	As a Market Strategy Director, I can view the sales trend by category from 2021 to 2025 so that I can spot high-growth segments.	Line chart shows the yearly trend; the insight line names the fastest-growing segment	High	Sprint-2
	Customer Satisfaction	USN-3	As a Market Strategy Director, I can compare customer ratings across categories so that I can judge satisfaction by segment.	Rating chart shows the average rating per category and updates with filters	Medium	Sprint-2
BI Manager (David Chen)	Manufacturer Performance	USN-4	As a BI Manager, I can view manufacturer market share and top manufacturers by sales so that I can see who leads.	Doughnut chart (top 7 + Others) and ranked bar chart are displayed	High	Sprint-1
	Manufacturing Footprint	USN-5	As a BI Manager, I can view sales by manufacturing country so that I can see where the market is concentrated.	Bar chart shows sales per country and updates with filters	Medium	Sprint-2
	Battery Mix	USN-6	As a BI Manager, I can view the battery technology share of sales so that I can track the battery mix.	Chart shows the sales share of LFP, NMC, NCA, Sodium-Ion and Solid-State	High	Sprint-1
Mobility Researcher (Dr. Sarah Williams)	Battery & Range Analysis	USN-7	As a Mobility Researcher, I can compare average driving range by battery type so that I can study battery performance.	Bar chart shows sales-weighted average range for each battery type	High	Sprint-2
		USN-8	As a Mobility Researcher, I can compare average range by vehicle category so that I can see range improvements across segments.	Bar chart shows average range per category in km	Medium	Sprint-2
All dashboard users	Filters & Exploration	USN-9	As a user, I can filter by category, battery, manufacturing country and manufacturer so that I can focus on the data I need.	KPIs, charts and the model table update immediately after each filter change	High	Sprint-1
		USN-10	As a user, I can reset all filters with one click so that I can return to the full view.	All four filters return to 'All' and the full data is shown	Medium	Sprint-1
		USN-11	As a user, I can sort the model explorer table by any column so that I can rank and compare models.	Clicking a header sorts the table and shows an up or down arrow	Medium	Sprint-2
	Persona Views & KPIs	USN-12	As a user, I can switch between the Strategy, BI and Research tabs so that I see the charts relevant to my role.	Each tab shows its persona description and its own set of charts	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
		USN-13	As a user, I can see KPI cards and an auto-generated insight line so that I get the key findings quickly.	Cards show sales, YoY growth, average range, rating, and models and makers	High	Sprint-1
Dashboard Administrator	Data Management	USN-14	As an Administrator, I can replace the sample dataset with a real EV dataset so that the dashboard reflects actual market data.	Updating the data array refreshes all KPIs, charts and the table without other code changes	Medium	Sprint-3